Event Boundaries and Recognition Memory

An Analysis using the Mnemonic Similarity Task (MST)

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1. Introduction

Human memory does not function as a continuous recording of experience. Instead, cognitive systems organize experiences into structured units called **events**. According to **event segmentation theory**[1], people naturally divide ongoing experiences into meaningful segments separated by **event boundaries**, such as changes in activity, context, or environment. These boundaries help structure information during encoding and influence how memories are later retrieved.

Understanding how event boundaries influence memory formation is an important topic in cognitive psychology and neuroscience. One widely used experimental paradigm for studying recognition memory and memory precision is the **Mnemonic Similarity Task (MST)**.[2]

The MST measures both **basic recognition memory** and **mnemonic discrimination**, which refers to the ability to distinguish highly similar stimuli from previously seen items. This discrimination ability is believed to depend on **pattern separation processes in the hippocampus**[3], which help encode overlapping experiences as distinct memory representations.

In the MST, participants first complete an **encoding phase**, during which they view a sequence of images. Later, in a **recognition phase**, they classify images as:

- **Old** – exact repetition of a previously seen image
- **Similar** – visually similar but not identical image
- **New** – completely novel image

Performance in the MST is typically summarized using two bias-corrected metrics:

Recognition Memory Index (REC)

REC = P(old | target) − P(old | foil)

This measure reflects the ability to correctly recognize previously studied items while correcting for response bias.

Lure Discrimination Index (LDI)

LDI = P(similar | lure) − P(similar | foil)

This measure captures **mnemonic discrimination**, reflecting the ability to distinguish perceptually similar stimuli from previously studied items.

The present study analyzes behavioral data from an MST experiment designed to investigate how **event boundaries and stimulus similarity influence recognition memory performance**.
Through exploratory data analysis and hypothesis testing, this study aims to better understand how recognition memory and pattern separation operate during the MST task.

2. Dataset and Experimental Design

The dataset contains behavioral responses from **160 participants**, divided into three experimental conditions:

- **Item-only condition (n = 56)**
- **Both condition (n = 51)**
- **Task-only condition (n = 53)**

Each participant completed two phases of the experiment: an **encoding phase** and a **recognition test phase**.

2.1 Encoding Phase

During the encoding phase, participants viewed a sequence of images organized into **40 events**, with each event containing **7 items**.
The structure of each event was:

- **Item 1 → Pre-boundary**
- **Items 2–6 → Mid-event**
- **Item 7 → Post-boundary**

Participants were asked to perform a semantic judgment task for each image:

"Will this object fit inside a shoebox?"

This task ensured that participants processed each stimulus meaningfully rather than passively observing the images.

2.2 Test Phase

During the recognition phase, participants classified each image as:

- **Old (o)** – exact repeat of an encoded image
- **Similar (s)** – visually similar but not identical image
- **New (n)** – image not shown during encoding

The recognition test contained:

- **60 target images**
- **60 lure images**
- **30 foil images**

Targets correspond to exact repetitions of encoded images, lures are visually similar variations of encoded items, and foils are completely new images.

2.3 Lure Similarity Bins

Lure images were categorized into **five similarity bins** based on their perceptual similarity to the original encoded image:

- **Bin 1 → Most similar**
- **Bin 2**
- **Bin 3**
- **Bin 4**
- **Bin 5 → Least similar**

This categorization allows researchers to examine how recognition performance changes as **stimulus similarity decreases**.

3. Methods

3.1 Data Processing

Each participant contributed two CSV files:

1. **Encoding file** (task phase)
2. **Recognition test file**

Stimulus categories were identified using filename conventions:

- **Targets** end with a.jpg
- **Lures** end with b.jpg
- **Foils** begin with foil/

Response times were extracted from the recorded reaction time variables in the CSV files.

For each participant, the following metrics were calculated:

- Recognition Memory Index (**REC**)[2]
- Lure Discrimination Index (**LDI**)[2]
- Probability of responding "Similar" for each lure similarity bin

3.2 Exploratory Data Analysis

Initial exploratory analysis was conducted to understand participant response patterns and verify that the dataset behaved as expected.

The following patterns were examined:

- Response distributions across stimulus types
- Recognition memory performance across experimental conditions

- Mnemonic discrimination across similarity bins
- Reaction time distributions

These analyses provide descriptive insight into the dataset before performing formal statistical testing.

3.3 Hypothesis Formulation and Statistical Tests

3.31 Hypothesis 1 — Recognition Memory Above Chance

This hypothesis tests whether participants demonstrate recognition memory and mnemonic discrimination above chance levels.

Null Hypothesis (H_0)
REC = 0 and LDI = 0

Alternative Hypothesis (H_1)
REC > 0 and LDI > 0

To evaluate this hypothesis, a **one-sample t-test** was used to determine whether mean REC and LDI values differ significantly from zero.

Before applying parametric tests, **normality was evaluated using Q-Q plots and the Shapiro–Wilk test**.

3.32 Hypothesis 2 — Effect of Lure Similarity

This hypothesis examines whether mnemonic discrimination performance varies depending on lure similarity.

Null Hypothesis (H_0)

P(similar | lure) is equal across similarity bins.

Alternative Hypothesis (H_1)

P(similar | lure) differs across bins and is lower for highly similar lures.

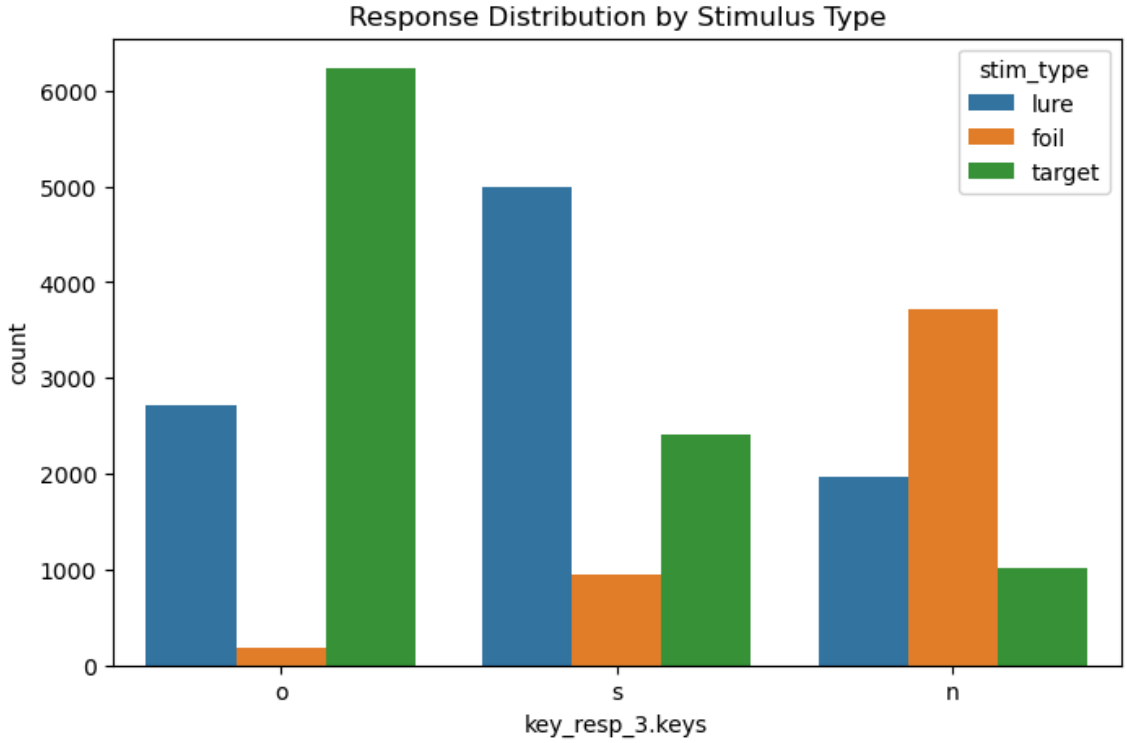
Because discrimination values are proportions bounded between 0 and 1, a **Kruskal-Wallis non-parametric test** was used to compare discrimination performance across similarity bins.

If significant differences were observed, **Dunn post-hoc comparisons with Bonferroni correction** were conducted.

4. Results

4.1 Response Distribution Across Stimulus Types

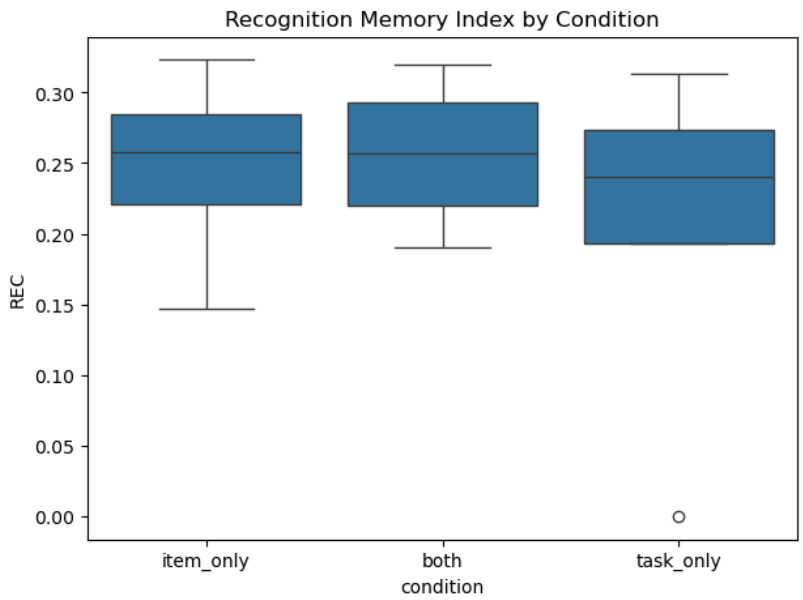
The first analysis examined how participants responded to different stimulus types.



Targets were most frequently classified as **Old**, lures were primarily labeled **Similar**, and foils were most often classified as **New**.
This response pattern indicates that participants correctly understood the recognition task and responded appropriately to different stimulus types.

4.2 Recognition Memory Performance

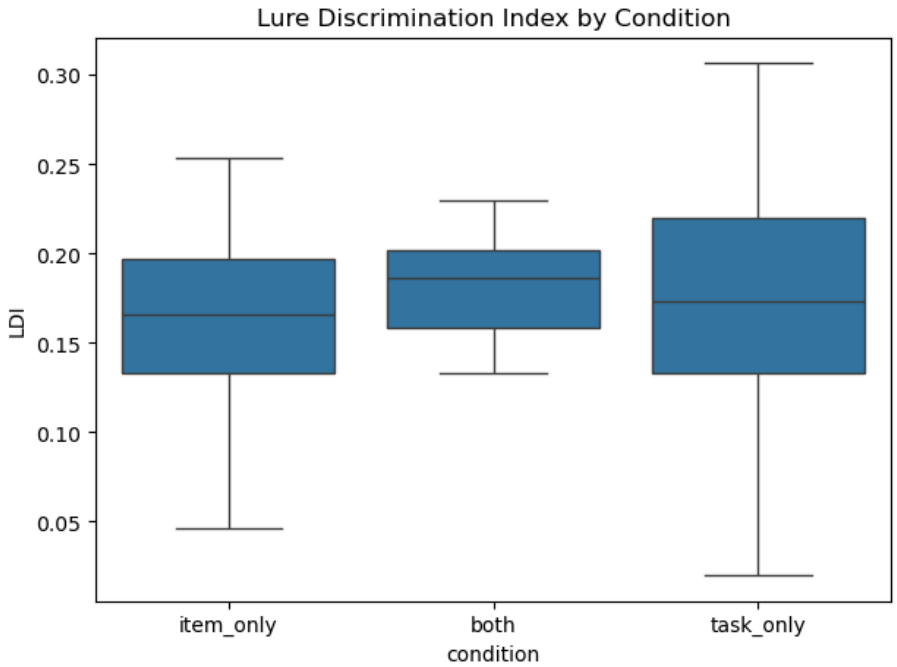
Recognition performance was evaluated using the **Recognition Memory Index (REC)**.



REC values were positive across all experimental conditions.
The mean REC value was approximately **0.25**, indicating strong recognition performance. Participants reliably distinguished previously studied items from novel stimuli, demonstrating effective memory encoding and retrieval.

4.3 Mnemonic Discrimination Performance

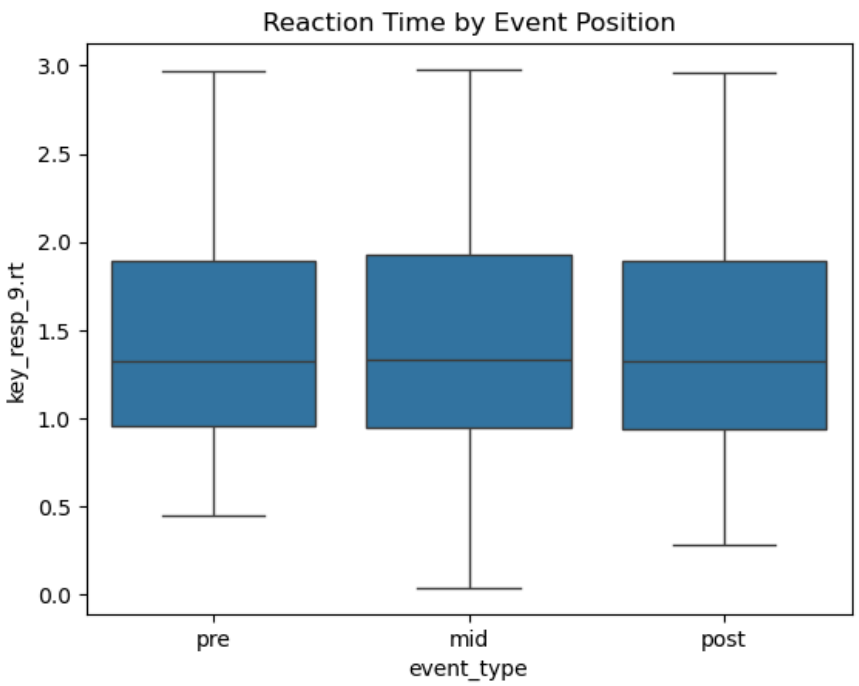
Mnemonic discrimination was evaluated using the **Lure Discrimination Index (LDI)**.



LDI values were positive across all conditions, with an average value of approximately **0.17**. Participants successfully discriminated similar stimuli from exact repetitions, indicating effective mnemonic discrimination.

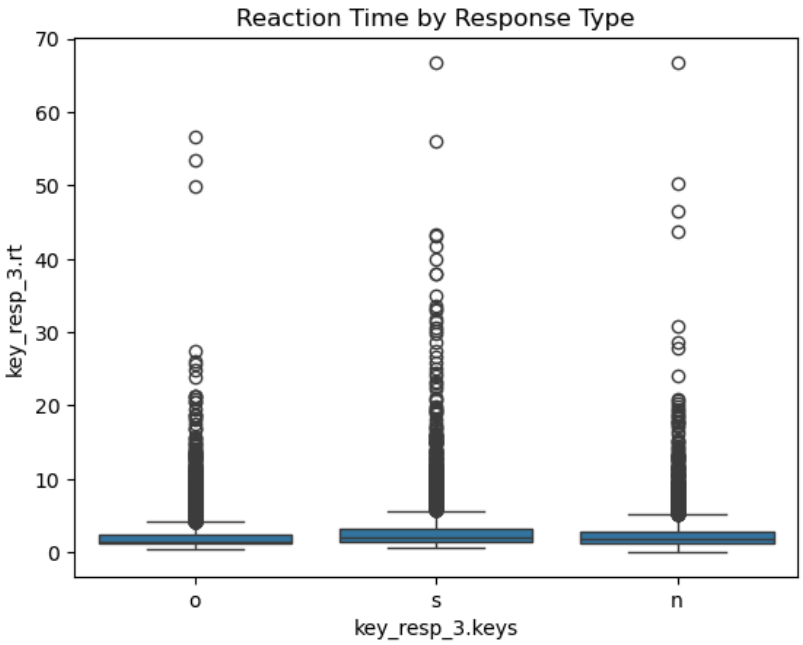
4.4 Reaction Time Analysis

4.41 Reaction Time by Event Position



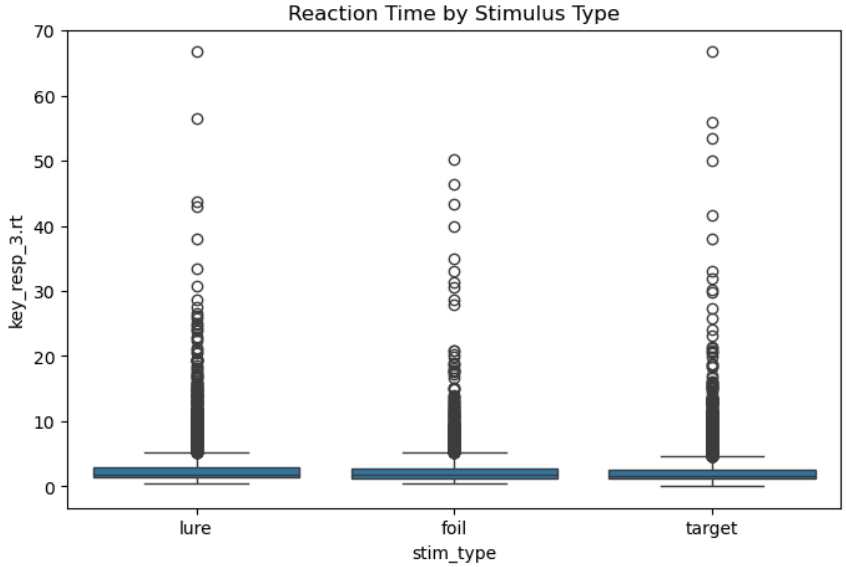
Reaction time distributions were similar across pre-boundary, mid-event, and post-boundary items. Event boundaries did not strongly influence encoding response speed in this dataset.

4.42 Reaction Time by Response Type



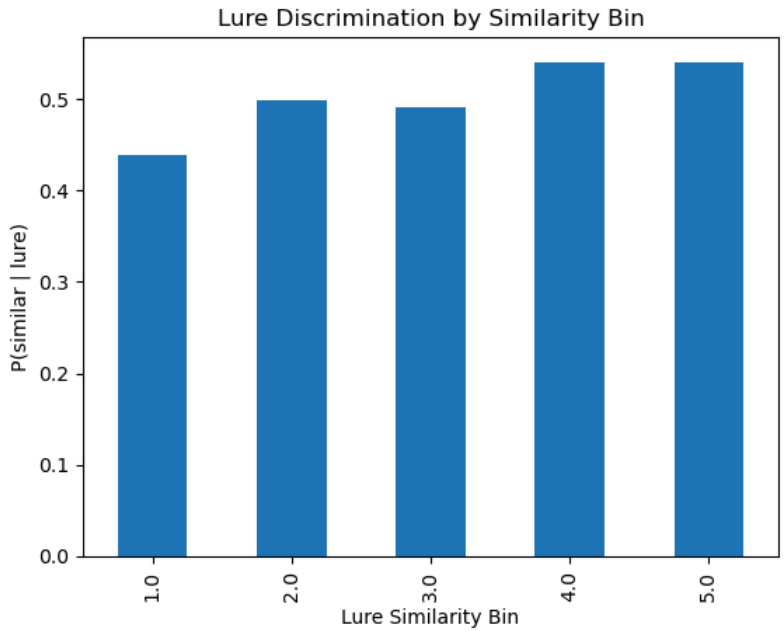
Responses labeled **"Similar"** showed slightly longer reaction times compared to "Old" and "New" responses.
This suggests that identifying perceptually similar items requires additional cognitive processing.

4.43 Reaction Time by Stimulus Type



Reaction time distributions are broadly similar across targets, lures, and foils.
Lure responses show slightly greater variability, suggesting that identifying similar items requires additional cognitive processing compared to recognizing old or new items.

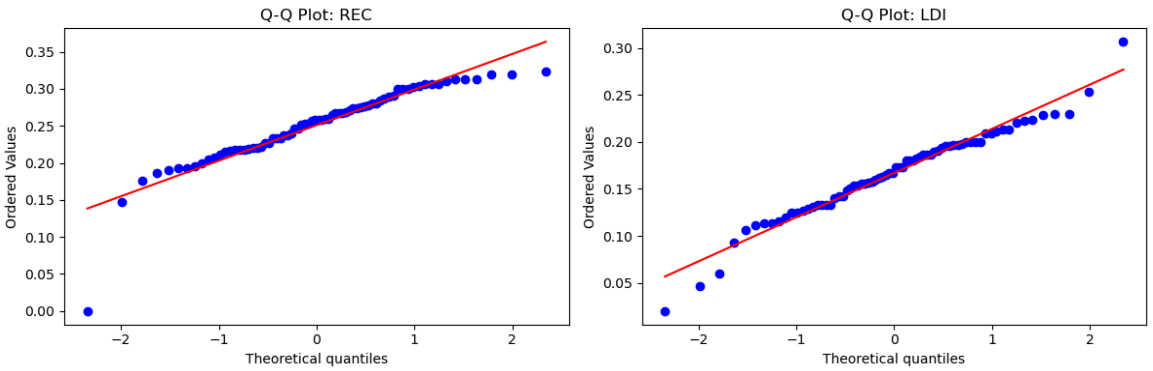
4.5 Pattern Separation Across Lure Similarity Bins



The probability of responding **"Similar"** increases from Bin 1 (most similar) to Bin 5 (least similar).
Highly similar lures are harder to distinguish from targets, while less similar lures are easier to identify. This pattern reflects **graded pattern separation**, where discrimination improves as similarity between stimuli decreases.

4.6 Hypothesis 1 Results — Recognition Memory Above Chance

First, the distributions of REC and LDI values were examined.



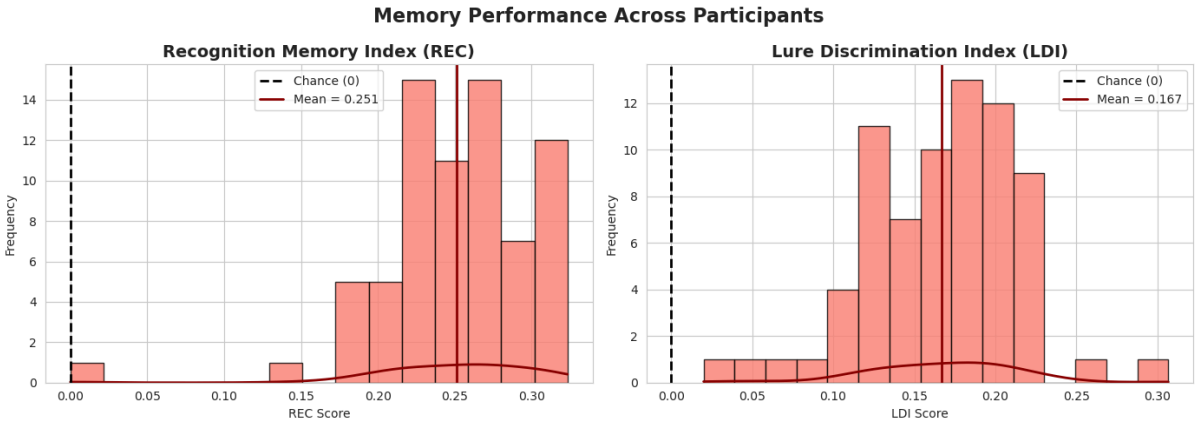
The Q-Q plots show that both REC and LDI values follow an approximately normal distribution with minor deviations.

The Shapiro–Wilk test results were:

REC: $W = 0.878$, $p < 0.001$
LDI: $W = 0.973$, $p = 0.118$

Although REC shows minor deviation from normality, both distributions are sufficiently close to normal to justify parametric testing.

Next, the distributions of REC and LDI values were visualized.



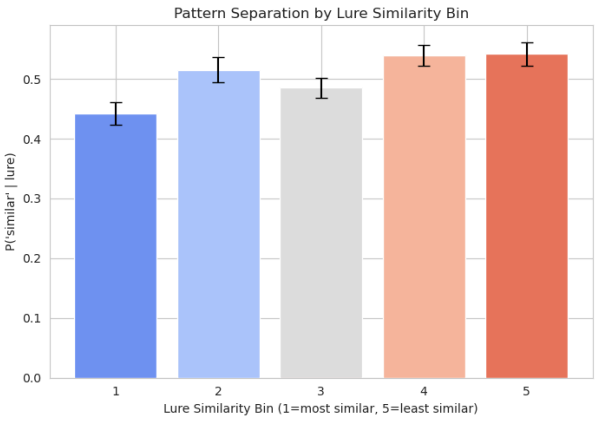
The dashed vertical line indicates chance performance (0), while the solid vertical line represents the mean score.

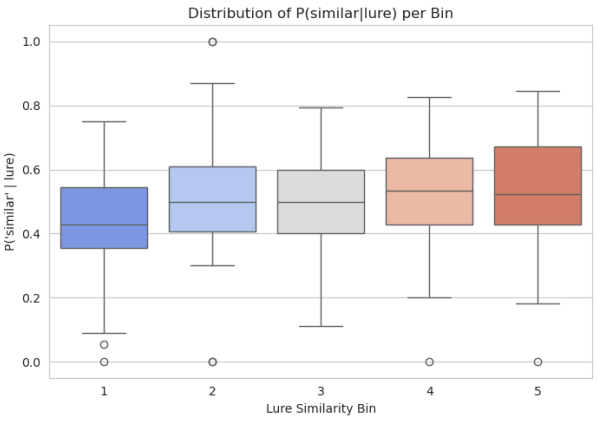
Observed values:

Mean REC ≈ **0.25**
Mean LDI ≈ **0.17**

A **one-sample t-test** confirmed that both REC and LDI values were significantly greater than zero. These results indicate that participants successfully recognized previously studied items and discriminated similar stimuli from targets. Therefore, the null hypothesis is rejected.

4.7 Hypothesis 2 Results — Lure Similarity Effects





The probability of responding "Similar" increases as lure similarity decreases.
Highly similar lures (Bin 1) were more difficult to identify, whereas less similar lures (Bin 5) were easier to distinguish.
A **Kruskal-Wallis test** revealed significant differences in discrimination performance across similarity bins.
Post-hoc **Dunn tests** indicated that discrimination performance for **Bin 1 was significantly lower than for Bin 5**, supporting the hypothesis that mnemonic discrimination decreases as stimulus similarity increases.

5. Discussion

The results demonstrate strong recognition memory and mnemonic discrimination across participants.

Both REC and LDI values were significantly above chance levels, indicating that participants successfully encoded and retrieved information during the MST task.

The similarity bin analysis revealed a graded pattern of mnemonic discrimination. Highly similar lures were more difficult to distinguish from targets, while less similar lures were easier to identify. This finding supports theoretical models of **hippocampal pattern separation**[3], where the brain distinguishes overlapping memories as separate representations.

Reaction time patterns further support this interpretation. Decisions involving similar stimuli required slightly longer response times, suggesting increased cognitive processing during mnemonic discrimination.

6. Limitations

The current analysis primarily relies on descriptive statistics and basic hypothesis testing.

Reaction time outliers were not removed, which may influence variability in response time distributions.

Future analyses should also explore interactions between **event boundaries and lure similarity**, which may provide further insight into how memory encoding processes operate in dynamic contexts.

7. Future Work(Planned Analysis)

Future work will extend the current analysis using more advanced statistical techniques.

Planned analyses include:

- **ANOVA comparing REC and LDI across experimental conditions**
- **Mixed-effects regression models examining event position effects**
- **Signal detection analysis using sensitivity (d′)**
- **Encoding–retrieval correlations using reaction time measures**
- **Analysis of memory performance across event positions to test whether event boundaries influence recognition memory(Hypothesis 3)**

These analyses will provide deeper insight into how event segmentation and stimulus similarity interact to influence recognition memory.

8. Conclusion

This study analyzed recognition memory and mnemonic discrimination using the Mnemonic Similarity Task.

Participants demonstrated strong recognition performance and successfully discriminated between similar stimuli. Memory performance varied systematically with lure similarity, supporting theoretical models of graded pattern separation.

These findings highlight the effectiveness of the MST paradigm for studying memory fidelity and provide a strong foundation for more advanced analyses in future work.

9. References

[1] https://pmc.ncbi.nlm.nih.gov/articles/PMC2852534/
[2] https://pmc.ncbi.nlm.nih.gov/articles/PMC3675184/
[3] https://pmc.ncbi.nlm.nih.gov/articles/PMC3183227/

Link to github(code)- https://github.com/Navdha1801/BRSM_project_phase1/tree/main

Link to MST dataset
-https://drive.google.com/drive/folders/10hYwuHbipr78Drb9QN3WSdT-Upspe-wL

10. Contribution

1. Navdha Bansal- Visualizations and Analysis
2. Paridhi Jain- Hypothesis 1 and Analysis
3. Gargie Tambe- Hypothesis 2 and Analysis